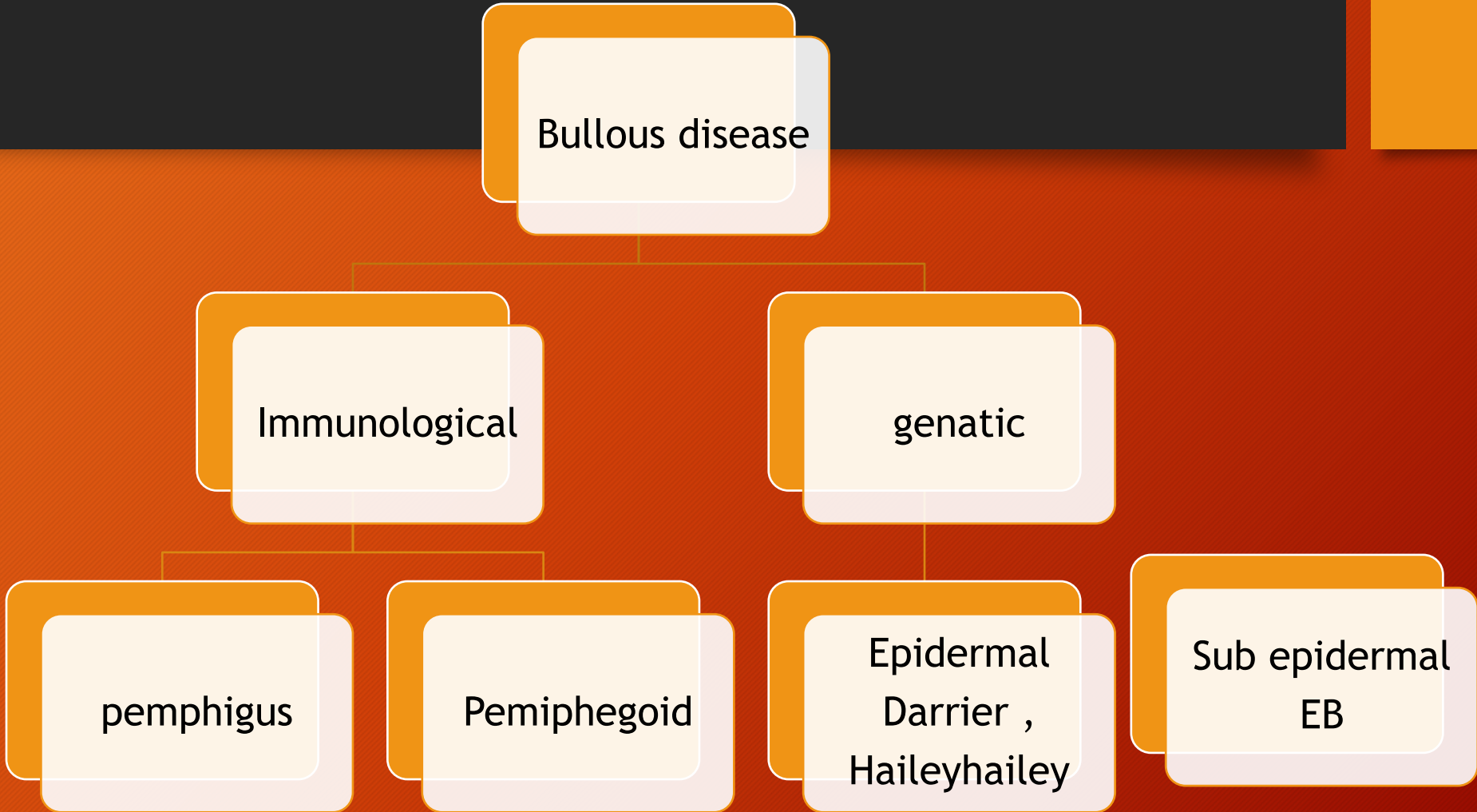


Genetic bullous disease

Dr eman gomaa
Consultant dermatologist

Bullous disease



```
graph TD; A[Bullous disease] --> B[Immunological]; A --> C[genetic]; B --> D[pemphigus]; B --> E[Pemiphegoid]; C --> F["Epidermal Darrier , Haileyhailey"]; C --> G["Sub epidermal EB"]
```

The diagram is a hierarchical flowchart. At the top is a box labeled 'Bullous disease'. A line from this box branches into two boxes: 'Immunological' on the left and 'genetic' on the right. From 'Immunological', a line branches into two boxes: 'pemphigus' and 'Pemiphegoid'. From 'genetic', a line branches into two boxes: 'Epidermal Darrier , Haileyhailey' and 'Sub epidermal EB'. Each box is a light cream color with a thin orange border and a slight drop shadow, set against a dark orange background.

Immunological

genetic

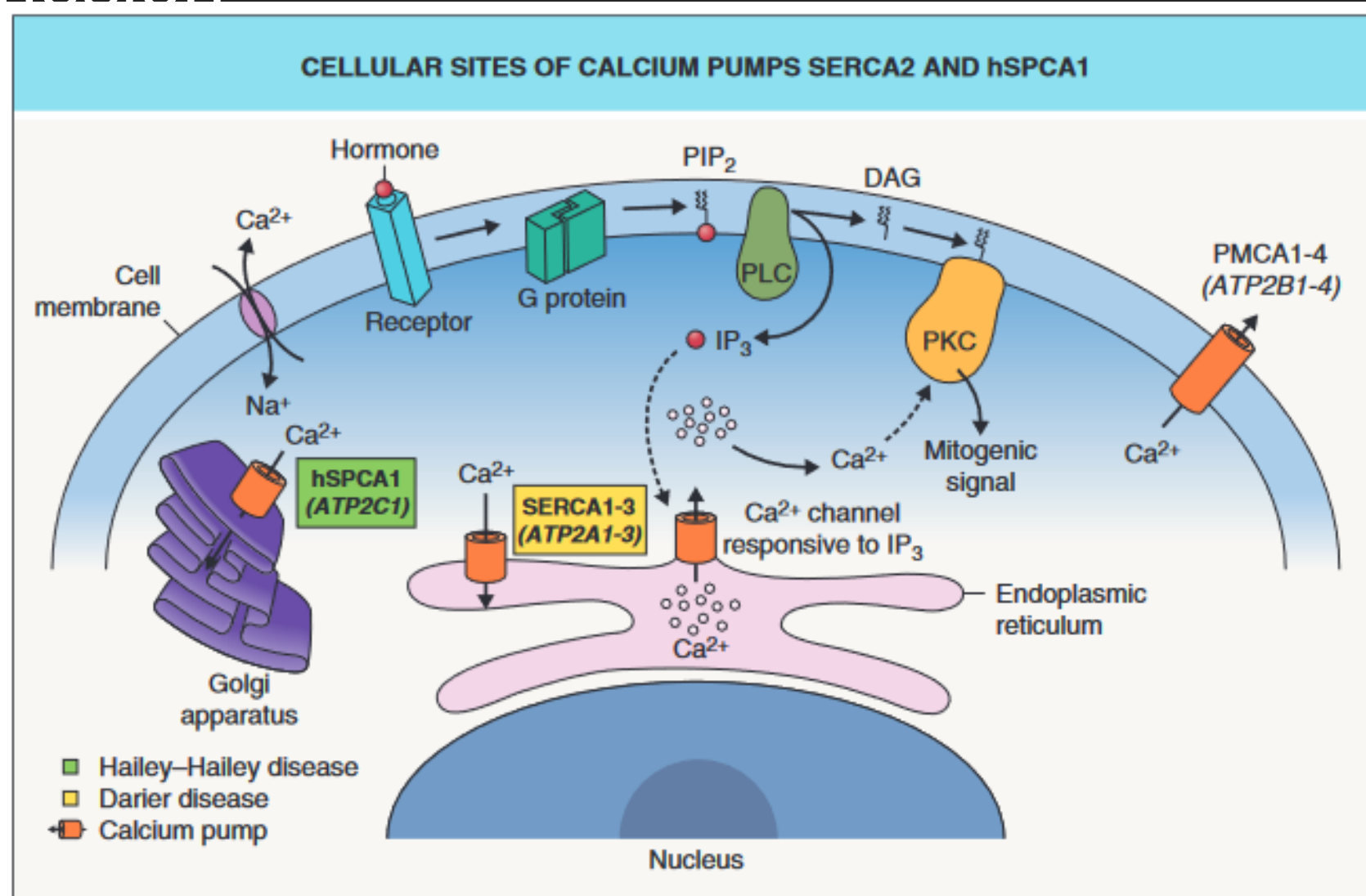
pemphigus

Pemiphegoid

Epidermal
Darrier ,
Haileyhailey

Sub epidermal
EB

Role of calcium (Ca^{2+}) in keratinocytes adhesion



- desmosomal proteins are synthesized in association with the endoplasmic reticulum (ER) (where they are folded and processed) then transported to the Golgi apparatus for further processing before being transported to the cell membrane
- The translation, translocation, folding, processing, maturation and transport of desmosomal proteins are regulated by the levels of luminal Ca^{2+} and Mn^{2+} in the ER and the Golgi apparatus.

- In keratinocytes, there are two calcium pumps
- 1. The sarco(endo)plasmic reticulum Ca^{2+} ATPase type 2 (SERCA2) that catalyses the transport of calcium (Ca^{2+}) from the cytosol into the lumen of the endoplasmic reticulum (ER).
- 2. The human secretory pathway $\text{Ca}^{2+}/\text{Mn}^{2+}$ ATPase protein 1 (SPCA1) that is found in the Golgi apparatus and supplies it with both Ca^{2+} and manganese (Mn^{2+})

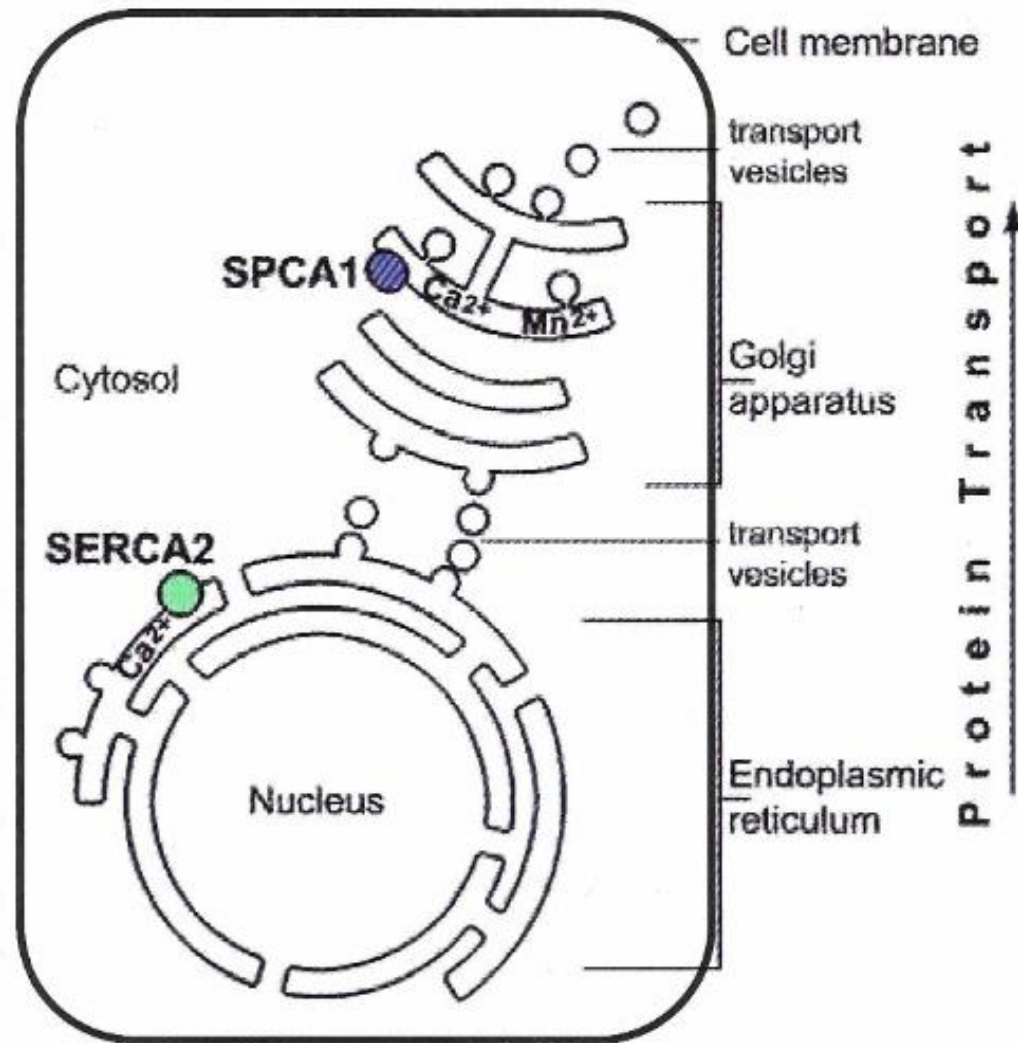


Fig 16.1 Localization of SERCA2 and SPCA1 in keratinocytes.

- Both Darier and Hailey-Hailey diseases are inherited (AD) and are caused by mutations in **Ca²⁺ pumps** inside keratinocyte
- darier disease is caused by mutations in ATP2A2 gene encoding SERCA2 in the ER.
 - Hailey-Hailey disease is caused by mutations in ATP2C1 gene encoding SPCA1 in the Golgi apparatus

- both diseases, defective Ca^{2+} filling of ER and Golgi apparatus >>>>
- - 1) Impairment of desmosome assembly and transport >>>>acantholysis.
- - 2) Accumulation of unfolded (unprocessed) protein >>> apoptosis (dyskeratosis)

Darrier disease

- AD >> ATP2A2 >>> SERCA2
- 6-20 years >>> increase in puberty , , , , increase in summer
- Exacerbation factor :
 - Summer
 - Uv
 - Sweating , heat , occlusion
 - Friction , 2ry infection
 - Lithium
 - Premenstrual , ocp

Clinical feature

- Symptom :itching , disfiguring , malodorous
- Skin lesion :
 - Papules :
 - Seborrheic area (keratotic ,crusted , red , brown (dirty ,warty papules)
 - Dorsum of hand , feet (flat topped papules (like plan wart)
 - Palmoplantar :papules , depression
 - Intertiginous area :mild fungating mass , malodorous
 - Guttate leukoderma



Fig. 59.2 Darier disease. Truncal involvement with a predilection for seborrheic areas. Keratotic papules can vary from red (A, B) to brown (C, D) in color and may become confluent. A & C, Courtesy, Julie V Schaffer, MD.



Fig. 59.3 Darier disease.
A Crusted papules at the hairline on the face and in the posterior auricular area. **B** Extensive facial involvement leading to ectropion.



Fig. 59.7 Darier disease. Multiple flat-topped papules on the dorsal aspect of the hand.



Fig. 59.14 Acrokeratosis verruciformis of Hopf. Involvement of the dorsal aspect of the foot with flat-topped papules.



59.8 Palmar involvement in Darier disease. **A** Both keratotic papules and keratin-filled depressions are seen. **B** Less commonly, a thick, spiked focal keratoderma is observed. *A, Courtesy, Kalman Watsky, MD. B, Courtesy, Julie V. Schaffer, MD.*





Fig. 59.9 Darier disease. Alternating longitudinal red and white streaks as well as notching of the free edge of the nail plate. *Courtesy, Antonella Tosti, MD.*

- Nail:
 - Longitudinal red , white line
 - V shaped notch
 - Subangual hyperkeratosis(wedg shape)
- Oral : 15-50%
 - Whitish papules >> palat



- 2 types :
 - Acute hemorrhagic type
 - Segmental type 1,2
- Complication :
 - Infection :fungal , bacterial , hsv (Kaposi varciliiform eruption)
 - Malignancy :SCC
 - Obstruction of salivary gland :painfull swelling
 - Neuropsychatric :epilepsy , mood disorder
 - Ocular :uveitis , ulceration

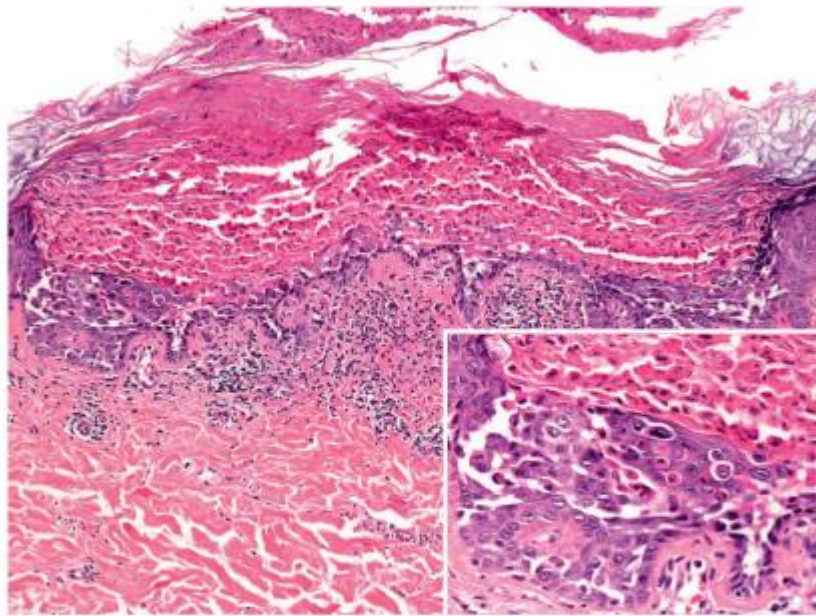


Fig. 59.13 Histopathology of Darier disease. Hyperkeratosis, grains and corps ronds (inset) are seen in addition to acantholysis leading to suprabasilar clefting. *Courtesy, Lorenzo Cerroni, MD.*

- H.p :
 - Acantholysis >>suprabasal
 - Dyskeratosis:nuclear condensation and perinuclear keratin clumping (due to apoptosis o f keratinocytes)
 - Corps ronds :acantholytic larg kcs in malipigian layer
 - grains :small , sc , sg
 - thickend epidermis , papilomatosis , hyperkeratosis

Dd



Fig. 59.14 Acrokeratosis verruciformis of Hopf. Involvement of the dorsal aspect of the foot with flat-topped papules.

- Sever SD
- Acrokeratosis verruciform of hopf
- Grovers disease
- HH, pemiphus vegitans



- General measures
- Light-weight clothes and sunscreen
- Daily skin care:
 - A Antimicrobial cleansers / washes
 - A Intermittent use of antibiotics and antifungal agents .
 - A Keratolytic emollients (to reduce scaling and irritation)

- Topical therapy
 - Topical retinoids
 - Topical 5-fluorouracil (1% or 5%) may be effective.

- 1) Systemic retinoids (isotretinoin and acitretin):
 - Advantages: very effective
- 2) Ocp
- 3) Cyclosporin
- 4) Surgical

Hailey-Hailey disease

Familial benign chronic pemphigus.

- AD , Mutations in the ATP2CA gene on chromosome 3q21 which encodes the Golgi-associated human secretory-pathway $\text{Ca}^{2+}/\text{Mn}^{2+}$ -ATPase isoform 1 (hSPCA1).
- SPCA1 dysfunction >> interfering with intracellular Ca^{2+} signaling A depletion of Ca^{2+} within the Golgi lumen >>> impaired processing of desmosomal proteins >>> acantholysis. Dyskeratosis is minimal or absent



Fig. 59.15 Hailey-Hailey disease. Erythematous, eroded plaque in the axilla. Note the intact flaccid vesicles at 2 and 7 o'clock.

- 2nd or 3rd decade
- HHD shows exacerbation during summer.
- Painful erosions and crusting
 - Pruritus (itching).
 - Malodor (psychosocial distress).
- flaccid vesicle on erythematous or normal-appearing skin —> ruptures easily —> macerated or crusted erosions
- Intertriginous sites
- Inframammary and vulvar lesions are common in women



Fig. 59.16 Hailey-Hailey disease. Chronic submammary lesion with erosions in a worm-eaten pattern.



Fig. 59.17 Hailey-Hailey disease. Circinate plaques on the back with erosions and crusting in the active borders.
Courtesy, Louis A Fragola, Jr, MD

- Nail
 - longitudinal white bands in the fingernails (longitudinal leukonychia)
- No mucosa

Complication :
Infection
Malignancy

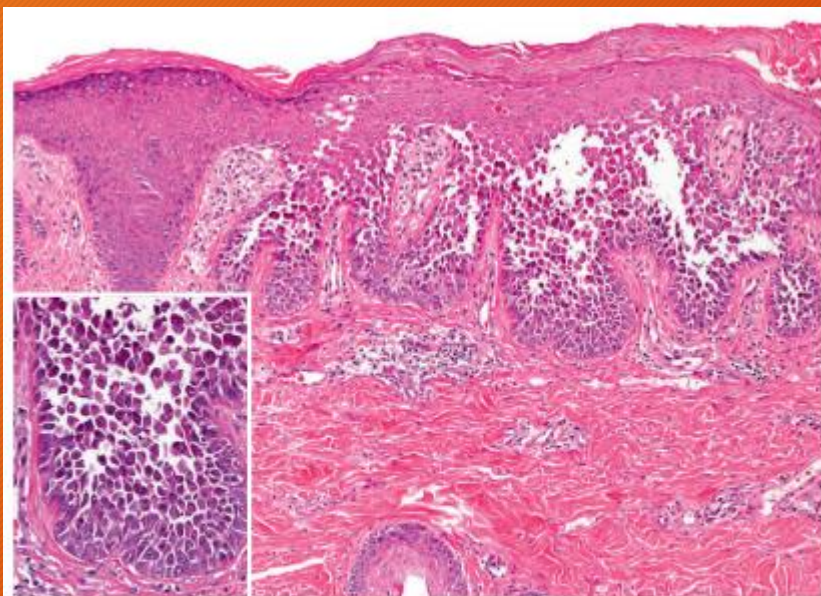


Fig. 59.18 Histopathology of Hailey–Hailey disease. Focal crusting and acantholysis beginning in the suprabasilar area and extending throughout the epidermis. The latter has been likened to a “dilapidated brick wall”. *Courtesy, Lorenzo Cerroni, MD.*

- Acantholysis WITHOUT dyskeratosis.
- Acantholysis:
 - o Supra-basilar cleft —»■ dermal papillae are lined by a single layer of basal cells and protrude into the blister cavities —▶ “villi”.
 - o More widespread within the epidermis,
 - o Incomplete or partial —> groups of acantholytic cells “dilapidated brick wall”

DD ??

Treatment

- Corticosteroids: effective treatment and are often combined with topical antimicrobials and cleansers
- botulinum toxin
- Oral retinoids are not clearly effective.

Oral antibiotic :tetracyclin

	Darier disease	Hailey-Hailey disease
Synonym	Darier-White disease, (Dys)Keratosis follicularis	Familial benign chronic pemphigus
Defect	ATP2A2 / SERCA2 (Endoplasmic reticulum)	ATP2CA / hSPCA1 (Golgi apparatus)
Age of onset	6 – 20 years (relatively early)	2 nd - 3 rd decade (relatively late)
Symptoms	Itching	Pain
Site	Seborrheic area	Intertriginous sites
Lesions	Keratotic papules	Vesicles, erosions, vegetations
Nail	Longitudinal red lines, V-shaped notches	Longitudinal white bands
Mucosa	Papules / palate	Rare
Complications	Infections Oral salivary glands Neuropsychiatric Ocular Malignant transformation	Infections Malignant transformation
Histopathology	Acantholytic dyskeratosis	Acantholysis, Minimal/no dyskeratosis
DD	Seborrheic dermatitis	Intertrigo
Treatment	General measures Topical Systemic Surgical	General measures Topical Surgical Systemic (least)

Epidermolysis bullosa

The sub-epidermal genetic bullous diseases

- group of clinically disorders(at least 30 phenotypes)
- with common features:
 1. Genetic transmission: ALL forms o f EB are inherited (AD or AR).
 2. Blister formation:
- 3. Mechanical fragility of the skin: development o f blisters following minor or insignificant trauma or traction to the skin.



Fig. 32.2 Localized epidermolysis bullosa simplex. A, B Bullae arising on the toes and plantar surfaces at sites of lateral or rotary traction. The majority of blisters occur in acral sites. B, Courtesy, Julie V. Schaffer, MD.

EBS

EB

EBS

EBD



Fig. 32.10 Generalized severe junctional epidermolysis bullosa. A Blisters on the elbow and large areas of denuded skin; note the bright red color in the axilla and groin. B Blisters and large erosions on the abdomen of an infant. B, Courtesy, Julie V. Schaffer, MD.



Fig. 32.12 Generalized severe recessive dystrophic epidermolysis bullosa. A Early proximal interdigital web formation together with atrophic scarring and loss of nails in a 5-year-old girl. B Partial "mitten" deformities of the hands in an older child. A, Courtesy, Julie V. Schaffer, MD.

- 1- EB simplex (EBS): within the epidermis. It's further subdivided into basal and supra-basal subgroups
-
- 2. Junctional EB (JEB): within the lamina lucida
-
- 3. Dystrophic E B (DEB): within the sub-lamina densa
- 4- kindler syndrome

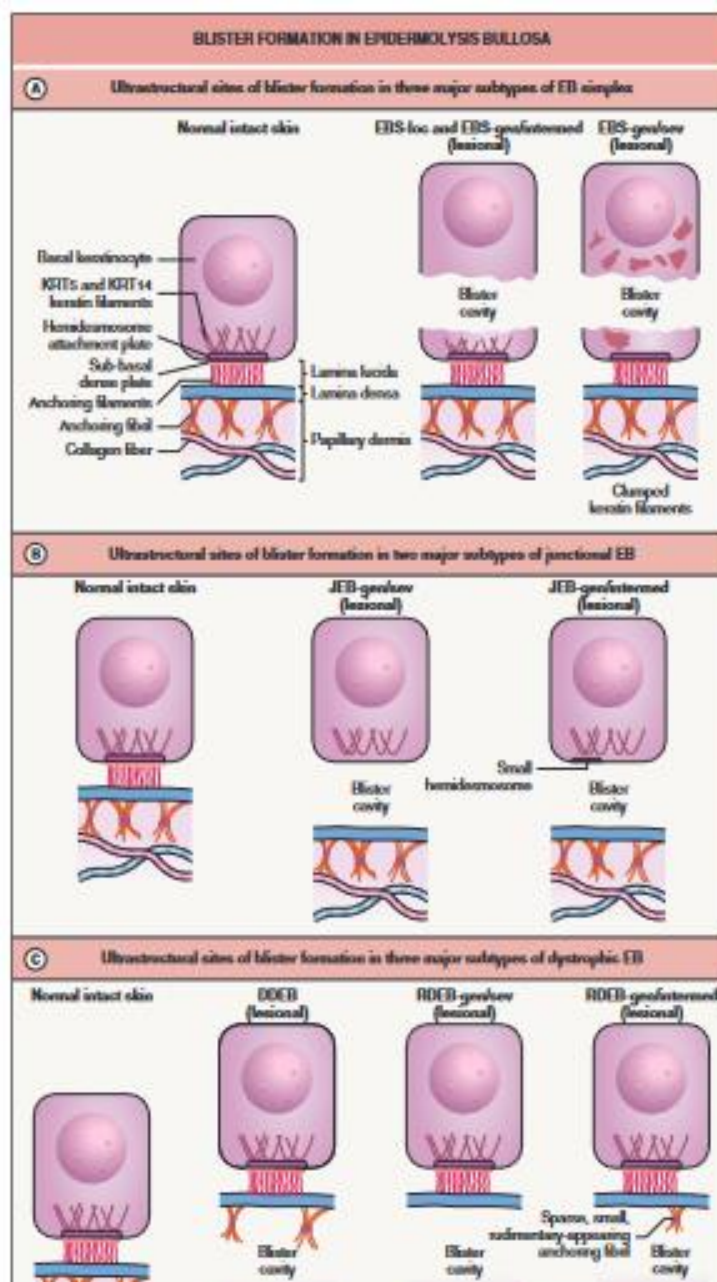


Fig. 32.1 Blister formation in epidermolysis bullosa (EB).

A Ultrastructural sites of blister formation in three major subtypes of EB simplex (EBS). The ultrastructure of normal intact skin is presented for comparison. In all three major subtypes of EB simplex – localized (EBS-loc), generalized intermediate (EBS-gen/intermed) and generalized severe (EBS-gen/sev) – blisters arise within the lowermost intracytoplasmic portion of the basal keratinocyte. In EBS-gen/sev, keratin filaments tend to coalesce into larger electron dense clumps, especially in lesional skin. Keratin filaments are absent or reduced in the rare autosomal recessive form of EBS, and in EBS with muscular dystrophy there is lack of integration of keratin filaments into the underlying hemidesmosome. Suprabasal forms of EBS that feature acantholysis (e.g. skin fragility syndromes) feature diminutive suprabasal desmosomes and perinuclear retraction of keratin filaments, while a higher cleavage plane between the granular and cornified cell layers is seen in EBS superficialis.

B Ultrastructural sites of blister formation in two major subtypes of junctional EB (JEB). All forms of JEB are characterized by blister formation within the lamina lucida of the dermal-epidermal junction. As such, the lamina densa remains firmly attached to the dermis which forms the base of the blister cavity. Sub-basal dense plates and anchoring filaments are absent in generalized severe JEB (JEB-gen/sev) and often attenuated in other subtypes of JEB. Hemidesmosomes are absent or extremely sparse and rudimentary in JEB-gen/sev, whereas they are normal or reduced in number and size in generalized intermediate JEB (JEB-gen/intermed).

C Ultrastructural sites of blister formation in three major subtypes of dystrophic EB (DEB). Blister formation occurs beneath the lamina densa in all forms of dystrophic EB. Anchoring fibrils appear normal in size and structure in dominant DEB (DOEB) skin, although they may be somewhat reduced in number. Consistent with the presence of sublamina densa cleavage, these anchoring fibrils remain attached to the blister roof, which is composed of intact epidermis and an attached basement membrane that includes the lamina densa. In contrast, anchoring fibrils are usually completely absent in lesional skin from patients with generalized severe recessive DEB (RDEB-gen/sev). In the generalized intermediate form of RDEB (RDEB-gen/intermed), anchoring fibrils are reduced in number and rudimentary in appearance.

Major form		Subtypes		Mode of inheritance	Defective protein(s)
		New name	Old / Other names		
EB simplex	Supra-basal	Superficialis		Unknown	
	Basal	Localized	Weber-Cockayne	AD	K5, K14
		Generalized intermediate	Koebner Generalized other Non- Dowling-Meara		
		Generalized severe	Dowling-Meara Herpetiformis		
		With mottled pigmentation with muscular dystrophy MCQ			
		With pyloric atresia (PA)		AR	Plectin MCQ
		Autosomal recessive	EBS-AR K14 EBS-AR P230 EBS-AR exophilin 5		Plectin $\alpha 6 \beta 4$ integrin K14 BPAG1 (BP230) Exophilin 5
Junctional EB	Generalized	Generalized severe	Herlitz MCQ	AR	Laminin-5 MCQ
		Generalized intermediate	non-Herlitz Other		Laminin-5 Collagen XVII
		With pyloric atresia MCQ			$\alpha 6 \beta 4$ MCQ integrin
	Localized	Localized			Collagen XVII
Dystrophic EB	Dominant	Generalized	Cockayne-Touraine and Pasini	AD	Collagen VII
		Localized (e.g. acral, pretibial, pruriginosa, nails only)			
	Recessive	Generalized severe	Hallopeau-Siemens		

Epidermolysis bullosa

EBS

AD
Weber cockayne
Koebner
Dowling Meara
ottled pigmentation

AR
Muscular
dystrophy

Pyloric
atrasia

Junctional EB

AD
Herlitz
Non herlitz
Pyloric
atrasia

Dystrophic EB

AD
Cockayne -
Touraine -
pasini

EB prurigenosa

AR
Hallopeau
siemens

Non hallopeau
siemens

KINDELLER

AR



Fig. 32.7 Acral peeling skin syndrome. Superficial peeling of the skin on the ankle and dorsal aspect of the foot. *Courtesy, E Sprecher, MD, PhD.*



Fig. 32.2 Localized epidermolysis bullosa simplex. **A, B** Bullae arising on the toes and plantar surfaces at sites of lateral or rotary traction. The majority of blisters occur in acral sites. *B, Courtesy, Julie V Schaffer, MD.*



Fig. 32.3 Generalized intermediate epidermolysis bullosa simplex. Extensive blistering on the buttocks (**A**) and blistering in association with focal keratoderma on the soles (**B**) of a 2-year-old girl. *Courtesy, Julie V Schaffer, MD.*



Fig. 32.8 Generalized severe epidermolysis bullosa simplex. Clustered vesicles in an arcuate array on the thigh of this child. *Courtesy, Julie V Schaffer, MD.*



Fig. 32.9 Generalized severe epidermolysis bullosa simplex. Diffuse palmar keratoderma in an adult.



Fig. 32.10 Generalized severe junctional epidermolysis bullosa. **A** Blisters on the elbow and large areas of denuded skin; note the bright red color in the axilla and groin. **B** Blisters and large erosions on the abdomen of an infant.
R. Courtesy, Julie V Schaller, MD.



Fig. 32.12 Generalized severe recessive dystrophic epidermolysis bullosa. **A** Early proximal interdigital web formation together with atrophic scarring and loss of nails in a 5-year-old girl. **B** Partial "mitten" deformities of the hands in an older child. *A, Courtesy, Julie V Schaffer, MD.*



Fig. 32.11 Dystrophic epidermolysis bullosa, pruriginosa. Prurigo-like papulonodules coalescing into linear plaques on the shin as a consequence of chronic scratching.



Fig. 32.4 Dominant dystrophic epidermolysis bullosa. **A, B** Erosions, scarring and milia on the fingers. Note the partial (**A**) to almost complete (**B**) loss of the nails. **C** A discrete area of scarring on the elbow with blisters, crusting and milia. *(Courtesy Julie V Schaffer, MD)*



Fig. 32.19 Kindler syndrome. **A** Poikiloderma of the face and neck with "skip" areas. **B** Erythema and atrophy on the dorsal hand. **C** Winkling due to atrophy and fusion between the fourth and fifth toes. This patient also had ectropion and urethral strictures. *C, Courtesy, Jean L. Bolognia MD.*

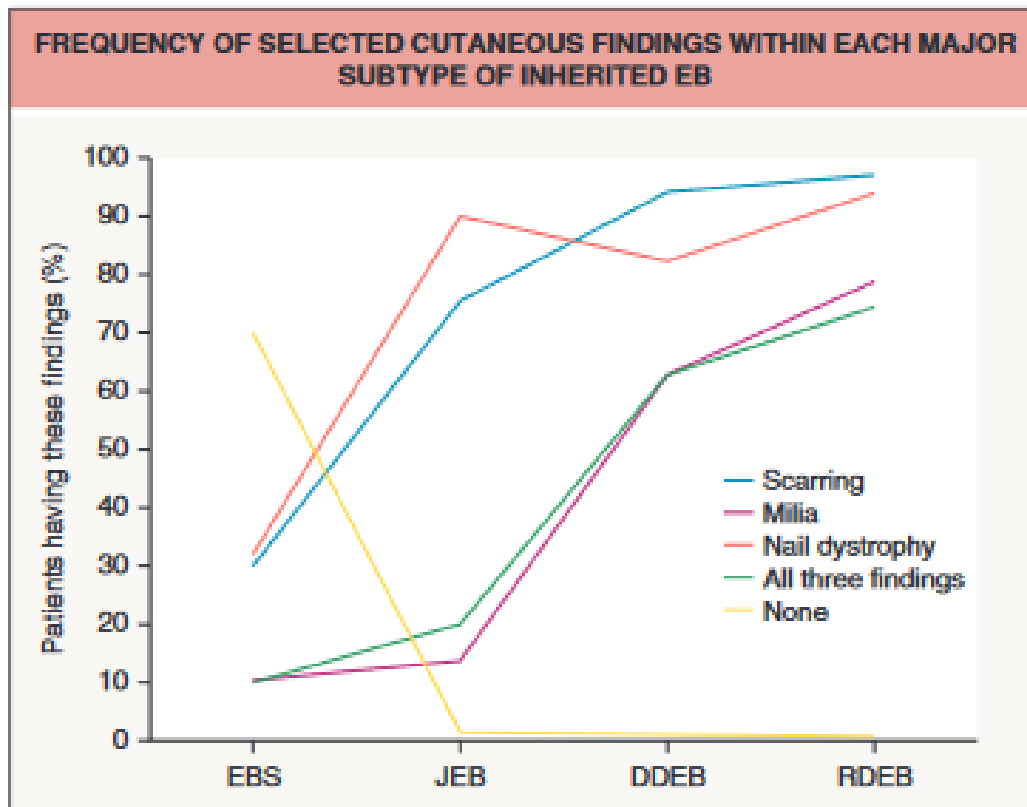


Fig. 32.5 Frequency of selected cutaneous findings within each major subtype of inherited epidermolysis bullosa (EB). Note the increasing frequency from localized EB simplex (EBS) to recessive dystrophic EB (RDEB). JEB, junctional EB; DDEB, dominant dystrophic EB.

Clinical features

- (1) Cutaneous Finding

General cutaneous features: all forms of inherited EB are characterized by

1. Mechanically fragile skin, skin blisters and erosions
2. Atrophic scarring
3. Dystrophic or absent nails, milia, and scarring alopecia of the scalp.

- Extra-cutaneous Findings (most frequently in RDEB)

- Eyes: corneal blisters, ulcers, scarring and ectropion
- Oral cavity: microstomia.
- Teeth: enamel hypoplasia, excessive caries and premature loss of teeth
- Respiratory: tracheolaryngeal stenosis (fatal)

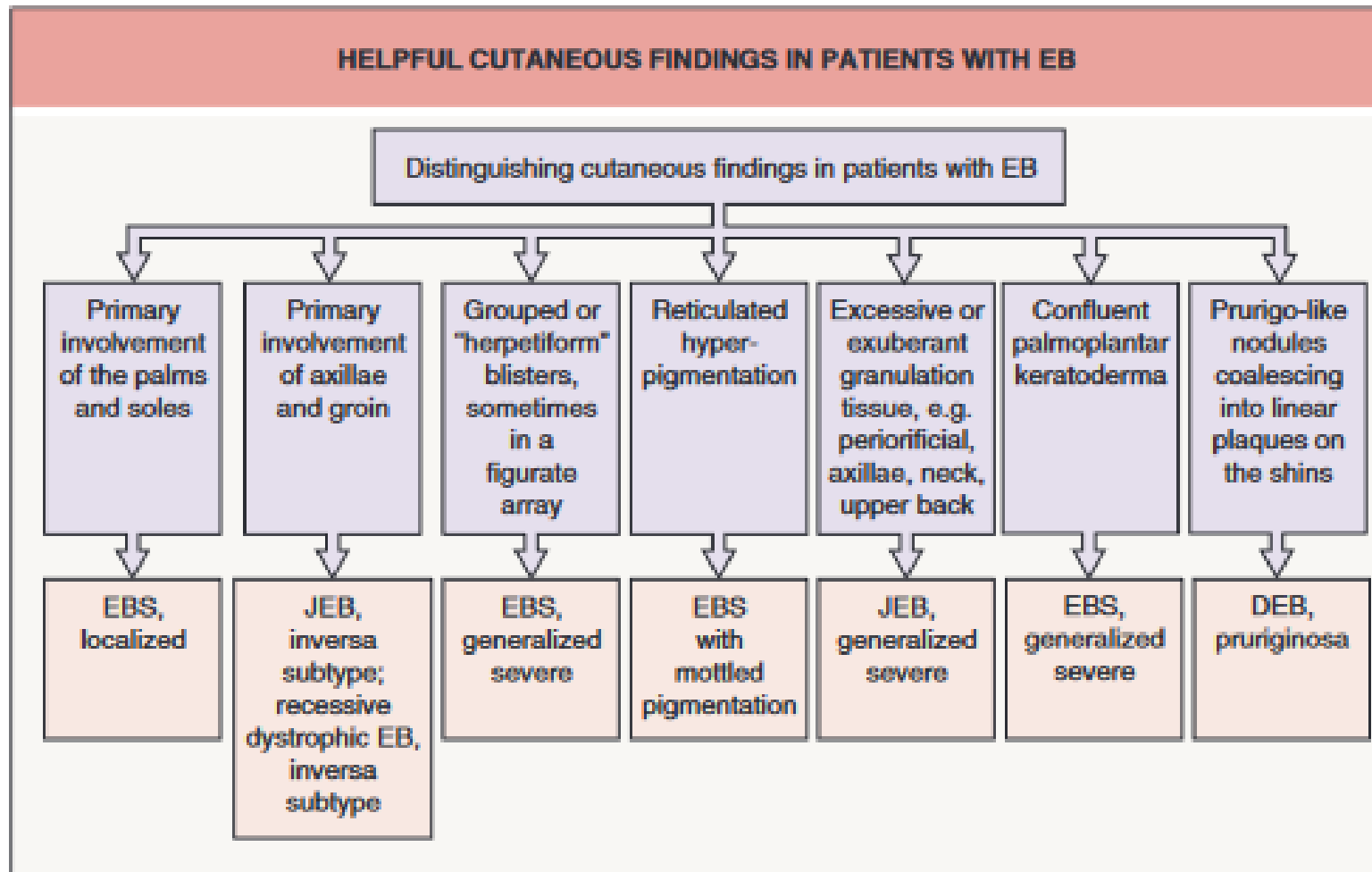
Gastro-intestinal: esophageal strictures, pyloric atresia, chronic

- Cardiac: dilated cardiomyopathy.
- Genitor-urinary: urethral meatal stenosis, hydroureter, .
- Musculo-skeletal: pseudosyndactyly, osteoporosis or osteopenia
- Bone marrow: severe anemia

MAJOR EXTRACUTANEOUS COMPLICATIONS OF EPIDERMOLYSIS BULLOSA		
Complication	EB subtype(s) most commonly affected	
	≥50% of patients	<50% of patients
<i>Eyes</i>		
Corneal blisters, ulcers and scarring	RDEB-gen/sev	JEB-gen/sev > RDEB-gen/intermed, RDEB-inv, JEB-gen/intermed
Ectropion formation		JEB-gen/sev; Kindler
<i>Oral cavity and upper airway (excluding blisters)</i>		
Microstomia	RDEB-gen/sev	JEB-gen/sev, RDEB-inv
Enamel hypoplasia	JEB (all subtypes)	
Excessive caries and premature loss of teeth	RDEB gen/sev, JEB-gen/sev	
Tracheolaryngeal stenosis	JEB-gen/sev	JEB-gen/intermed
<i>Gastrointestinal tract</i>		
Esophageal strictures	RDEB-gen/sev, RDEB-inv	RDEB-gen/intermed, Kindler > JEB-gen/sev
Pyloric atresia	JEB-PA, EBS-PA	
Malnutrition/failure to thrive	RDEB-gen/sev, JEB-gen/sev	JEB-gen/intermed
Severe constipation	RDEB-gen/sev	JEB-gen/sev, EBS-gen/sev
GERD	RDEB	JEB, EBS-gen/sev
Colitis		Kindler, RDEB-gen/sev

		sev
<i>Genitourinary tract</i>		
Urethral meatal stenosis		RDEB-gen/sev, JEB-gen/sev, Kindler
Chronic renal failure*		RDEB-gen/sev
Hydroureter and hydronephrosis		JEB-PA, JEB-gen/sev
Nephrotic syndrome	JEB-resp/renal	JEB-gen/sev
<i>Heart</i>		
Dilated cardiomyopathy		RDEB-gen/sev > JEB, RDEB-gen/intermed
<i>Musculoskeletal system</i>		
Pseudosyndactyly	RDEB-gen/sev	RDEB-gen/intermed, Kindler
Osteoporosis or osteopenia	RDEB-gen/sev	RDEB-gen/sev, JEB-gen/sev
Muscular dystrophy	EBS-MD	
<i>Bone marrow</i>		
Severe multifactorial anemia	RDEB-gen/sev, JEB-gen/sev	
*Causes include renal amyloidosis and glomerulonephritis.		

Specific cutaneous feature



EB type / subtype	Characteristic features	Distribution
EBS superficialis (EBSS)	Superficial peeling of the skin, no blistering	
EBS, localized		Palms and soles
EBS-generalized severe (Dowling–Meara “DM”)	Palmo-plantar keratoderma	Grouped (herpetiform) - arcuate or polycyclic
EBS with mottled pigmentation (EBS-MP)	Reticulated hyperpigmented macules	
JEB-H.	Excessive or exuberant granulation tissue	Symmetric - periorificial, axilla, neck, upper back
“Inversa” subtypes of JEB and RDEB		intertriginous areas (e.g. axillae, inguinal creases)
Pretibial DEB		Exclusively on the shins

Ivestogation

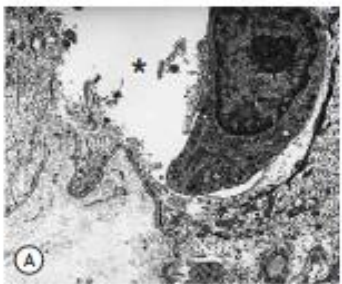
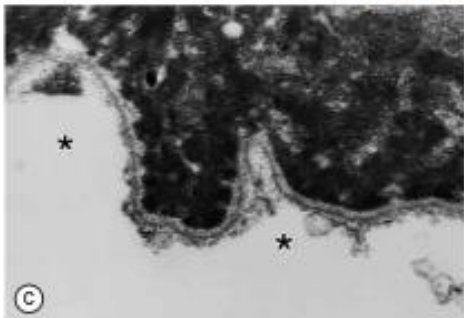
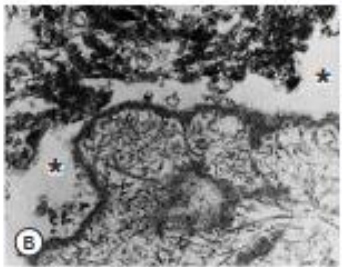
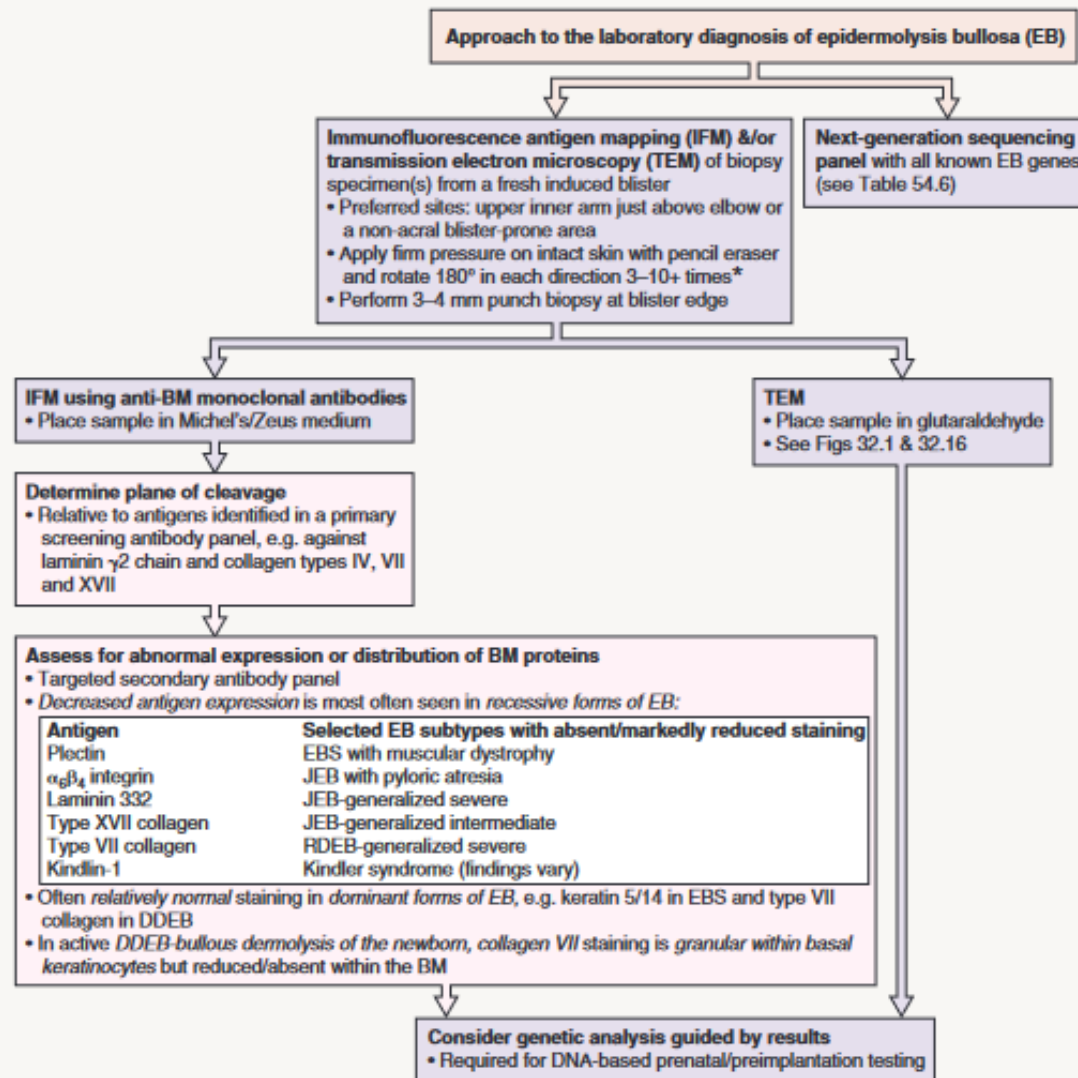


Fig. 32.16 Ultrastructure of epidermolysis bullosa simplex (EBS), junctional epidermolysis bullosa (JEB) and recessive dystrophic epidermolysis bullosa (RDEB). A Electron microscopy (EM) of an induced blister in localized EBS demonstrates skin cleavage (asterisk) within the inferior-most portion of the basilar keratinocyte. B EM of a spontaneous blister from a patient with the severe generalized JEB reveals skin cleavage (asterisks) within the lamina lucida. Hemidesmosomes, sub-basal dense plates and anchoring filaments are all absent. In contrast, anchoring fibrils are still present in normal amounts within the underlying dermis. C EM of the roof of a spontaneous blister from a patient with generalized severe RDEB reveals cleavage (asterisks) beneath the level of the lamina densa. Anchoring fibrils are absent along the epidermal roof of the blister.



- light microscopy: NO role in the diagnosis of EB.
-
- Transmission electron microscopy (TEM): distinguishes the four major EB types and can assess specific structures within the DEJ.
-
- Immunofluorescence antigenic mapping: determine the level of blister formation through staining the dermal-epidermal zone

APPROACH TO THE LABORATORY DIAGNOSIS OF EPIDERMOLYSIS BULLOSA (EB)



*May be more difficult to induce blisters in patients with localized EBS or DDEB

DD in neonates or young infants:

- Genetic diseases e.g. bullous congenital ichthyosiform erythroderma or incontinentia pigmenti (vesicular stage).
- Traumatic blisters e.g. sucking blisters.
- Infectious diseases e.g. herpes simplex, SSSS, bullous impetigo).
- Tumors e.g. bullous mastocytosis.
- Autoimmune blistering diseases e.g. Pemphigus or herpes gestationis, acquired transplacentally.
- Metabolic disorders e.g. acrodermatitis enteropathica.
- DD of EBSS: Peeling skin syndrome.

Treatment:

- NO specific treatment - preventive and symptomatic.
- 1. General preventive measures:
 - Prevention of mechanical trauma: padding over bony prominences, soft / loose-fitting clothing and shoes.
 - Prevention of infection: bathing or soaking with 0.005% sodium hypochlorite or 0.25% acetic acid, antibiotics.

- Skin dressing: non-adhesive dressing e.g. vaseline-impregnated gauze, soft silicone dressings

DRESSINGS FREQUENTLY USED IN PATIENTS WITH EPIDERMOLYSIS BULLOSA		
Dressing type	Use	Examples [®] , TM
Soft silicone dressings ("low-tack")	• Suitable as primary or secondary dressings in many types of EB • May have a foam backing for mechanical protection or to wick exudate away from the wound	Mepitel, Mepilex, Mepilex Transfer, Mepilex Border
Non-adherent lipido-colloid dressings	• Suitable for primary contact	Urgotul
Hydrogel dressings	• Provide moisture to drier wounds to facilitate healing • May provide relief from pain or itch	Flexigel, Curagel, ActiForm Cool
Foam dressings	• Useful to absorb moderate amounts of wound exudate	Mepilex, Allevyn
Absorptive dressings	• Useful to absorb heavy amounts of wound exudate	Eclipse, Sorbion Sana
Silver-containing dressings	• Clinically infected or highly colonized wounds • Avoid prolonged use due to risk of silver absorption	Urgotul SSD, Mepilex AG
Other antimicrobial dressings	• Clinically infected or highly colonized wounds	PolyMem, Cutimed Sorbact, Activon Tulle, Suprasorb X + PHMB

Table 32.4 Dressings frequently used in patients with epidermolysis bullosa (EB).

- Systemic therapies: e.g
- . Phenytoin (inhibits collagenase),
 - Tetracycline,
 - Thalidomide
 - and cyclosporine (DEB pruriginosa)
 - Systemic retinoids
- Future and trial therapies: e.g. gene therapy, Intra-dermal injection of human type VII collagen or fibroblasts, Bone marrow-derived stem cell transplantation.

Table 2-5: Epidermolysis Bullosa

EB Subtype	Inh	Gene	Clinical Features
EB SIMPLEX (EBS) Split: Epidermal Basal Layer			
Dowling-Meara (EBS Herpetiformis)	AD	K5/K14	Onset at birth, grouped or herpetiform blisters (figurate), significant mucosal membrane and laryngeal/esophageal involvement (± hoarseness) , nail dystrophy, confluent PPK, scarring, early death EM: clumped tonofilaments in basal keratinocytes
Weber-Cockayne (Localized)	AD	K5/K14	Onset typically childhood/adolescence, palmoplantar bullae/erosions, heal without scarring
Koebner (Generalized)	AD	K5/K14	Generalized bullae at birth, PPK, nail dystrophy, mucosal erosions, heals without scarring
EBS Muscular Dystrophy	AR	Plectin	Widespread bullae at birth, muscular dystrophy , scarring, hair/nail/tooth/oral disease, early death
EBS Mottled Digmmentation			Resembles localized and generalized EBS + reticulated hyperpigmentation over trunk
JUNCTIONAL EB (JEB) Split: Basement Membrane (Lamina Lucida)			
Herlitz (EB Lethalis) Premature termination codon	AR	Laminin 5 (laminin-332)	Severe, widespread bullae, nonhealing exuberant granulation tissue (perioral, axillae, neck), enamel defects, absent nails, mucosal involvement (respiratory/GI tract with hoarseness), early death
Non-Herlitz (Generalized Atrophic Benign EB or GABEB)	AR	Laminin 5 or BPAG2 (BP180)	Widespread bullae at birth, heal with atrophic scars , mild oral involvement, scarring alopecia , nail dystrophy, improves with time
JEB with Pyloric Atresia	AR	α6β4 integrin	Severe congenital blistering, hydronephrosis, pyloric atresia , mucosal erosions
DYSTROPHIC EB (DEB) Split: Dermal (Sublamina Densa)			
Hallopeau-Siemens Recessive DEB (RDEB-HS) Premature termination codon	AR	Type VII collagen	Severe widespread bullae at birth, heals with atrophic scarring (hands/feet → ' mitten deformity '), milia, mucosal strictures, nail dystrophy, ↑skin, oral, esophageal SCCs
Non-Hallopeau-Siemens (RDEB-nHS)	AR	Type VII collagen	Skin changes localized to acral bony prominences, Hallopeau-Siemens symptoms but less severe
Cockayne-Touraine (DDEB-CT)	AD	Type VII collagen	Bullae mainly over extremities, heal with milia/atrophic scars/keloids, nail dystrophy
Pasini Variant (DDEB-P)	AD	Type VII collagen	Similar to Cockayne subtype + albopapuloid lesions (white perifollicular papules, slowly enlarge)
Transient Bullous Dermolysis of the Newborn	AD/AR	Type VII collagen	Uncommon variant of dystrophic EB; benign, self-limited disease without scarring in newborn period

Aplasia Cutis Congenita (ACC) (Figure 2.2A, B)

- Onset before birth; localized defect in epidermis, dermis and/or fat; variable appearance, typically along midline
- Presents with erosion, ulceration, scar or membranous defect (ovoid lesion covered by an epithelial membrane)
- Hair collar sign (Figure 2.2C): ring of dark long hair encircling lesion; $\pm$ marker of underlying neural tube defect
- Typically isolated abnormality, but may be associated with developmental anomalies or following disorders:

Bart syndrome	ACC of lower extremities + epidermolysis bullosa (DDEB)
Adams-Oliver syndrome	ACC on scalp (with skull ossification defect) + CMTC + limb defects (reductions, syndactyly) + cardiac abnormalities
Seitles syndrome	Bilateral temporal ACC + abnormal eyelashes, 'leonine' facies, upward-slanting eyebrows



Figure 2.2:

A: ACC, cicatricial
(Courtesy of Dr. Michelle B. Bain)

B: ACC, ulcerated
(Reprint from Lipsker D. *Clinical Examination and Differential Diagnosis of Skin Lesions*. Paris, France: Springer France;2013.)

C: Hair collar sign in ACC*

D: CMTC in Adams-Oliver syndrome*

Other bullous disease

Bullosis diabeticorum

Coma blisters

Friction blisters

Bullous small vessel vasculitis

Bullous drug eruptions

Bullous insect bite reactions

Delayed postburn/postgraft blisters

Edema blisters

Additional vesiculobullous disorders



Fig. 33.1 Bullosis diabeticorum. A large, tense bulla overlying the medial malleolus. Note the absence of erythema or edema in the surrounding

Key features

- A rare condition associated with diabetes mellitus
- Tense blisters develop on normal-appearing skin in acral sites (feet, lower legs, hands); there is often an association with peripheral neuropathy
- Histologically, intraepidermal and/or subepidermal blisters are observed
- Spontaneous healing usually occurs in 2 to 6 weeks

Key features

- Associated with loss of consciousness, neurologic diseases, or prolonged immobilization
- Tense blisters develop on normal-appearing skin
- Histologically, intraepidermal or subepidermal blisters are observed, with a characteristic necrosis of sweat glands
- Spontaneous healing occurs within 1 to 2 weeks



Fig. 33.2 Coma bullae. Blisters developed in an area of pressure in a previously comatose patient.



Fig. 33.3 Neurologic blisters. Tense blisters on the dorsal aspect of the fingers on the hemiplegic side in a patient with a history of a cerebrovascular accident.

ADDITIONAL VESICULOBULLOUS DISORDERS

Disorder	Characteristic features
Bullous dermatitis artefacta (self-inflicted injury) ^{30,31}	• Can be produced by a thermal, electrical, or chemical burn as well as application of a vesicant, coolant, or “salt and ice” (see Ch. 88) • Histologic features vary depending upon etiology, but the most common pattern is a subepidermal blister with epidermal necrosis +/- dermal damage
PUVA-induced acrobullous dermatosis	• Sudden appearance of sterile, tense blisters, usually on the distal extremities • Due to weakening of epidermal–dermal cohesion plus friction or trauma • DDX: phototoxicity (more widespread) and PUVA-induced bullous pemphigoid
Fracture blisters	• Tense bullae that develop within edematous skin and overlie bony fractures • Most common sites: distal tibia, ankle, foot, elbow, wrist • Thought to be due to shearing forces related to the injury plus post-traumatic edema
Sucking blisters	• Seen in newborns in areas accessible to the infant’s mouth, e.g. forearm • Develop <i>in utero</i>



Fig. 33.8 Bullous insect bite reactions. The sterile blister fluid is serous and the ankle is a common site. *Courtesy, Luis Requena, MD.*

BULLOUS DRUG ERUPTIONS		
Disease	Characteristic features	Commonly implicated drugs
Fixed drug eruption (see Ch. 21)	- Sharply circumscribed erythematous to dusky violaceous patches - Central blisters or erosions may appear - Often resolves with postinflammatory hyperpigmentation - Recurrence at same location(s) following drug re-exposure	Sulfonamides, NSAIDs, tetracyclines, barbiturates, aspirin, acetaminophen (paracetamol), metronidazole, phenolphthalein
Stevens–Johnson syndrome (SJS), toxic epidermal necrolysis (TEN) (see Ch. 20)	- Prodrome of fever and painful skin - Areas of dusky erythema associated with epidermal detachment (varying from <10% [SJS] to >30% [TEN]) - Mucosal involvement	NSAIDs, antibiotics (sulfonamides and β -lactams), aromatic anticonvulsants, allopurinol
Drug-induced autoimmune blistering diseases (see Chs 29–31)	- Primarily, linear IgA bullous dermatosis (Fig. 33.7), bullous pemphigoid, pemphigus - Diagnosis based on histologic findings, immunofluorescence studies and drug history	- Linear IgA bullous dermatosis: vancomycin > β-lactams, captopril, NSAIDs - Bullous pemphigoid: diuretics (especially furosemide), antibiotics - Pemphigus: captopril, β-lactams, penicillamine, gold
Drug-induced pseudoporphyria (see Ch. 49)	- Eruption resembles porphyria cutanea tarda - Porphyrin determinations are within normal limits	NSAIDs (especially naproxen), nalidixic acid, thiazides, furosemide, tetracyclines
Acute generalized exanthematous pustulosis (see Ch. 21)	- Acute onset, usually occurring within 2 days of drug exposure - Areas of erythema studded with pustules; occasionally vesicles - Fever, malaise, leukocytosis	β -lactams, macrolides, pristinamycin, terbinafine, calcium channel blockers (diltiazem), hydroxychloroquine, carbamazepine, acetaminophen, metronidazole
Phototoxic drug eruptions (see Ch. 87)	- Limited to sun-exposed areas - Resembles exaggerated sunburn	Tetracyclines (especially doxycycline), quinolones, psoralens, NSAIDs, diuretics
Bromoderma and iododerma (see Ch. 21)	- Acneiform lesions, papulopustules, nodules, or even vegetating lesions simulating pemphigus vegetans - Clear or hemorrhagic blisters can develop (more common in iododerma)	Bromides, iodine-containing drugs (e.g. amiodarone), radiographic contrast media
Palmoplantar erythrodysesthesia (acral variant of toxic erythema of chemotherapy; see Ch. 21)	- Painful erythema develops primarily on the palms, soles and digits following chemotherapy administration - The skin becomes edematous, its color changes to dark red or violet, and blisters and erosions may develop	Cytarabine, doxorubicin, capecitabine, 5-fluorouracil (especially prolonged infusions), multikinase inhibitors (e.g. sorafenib, sunitinib), busulfan, taxanes, clofarabine, pralatrexate

33 A 73-year-old patient presents with a number of intact blisters on the legs and torso. The blisters measure up to 6 cm in diameter and the patient describes a preceding pruritic eruption. A skin biopsy is performed that shows a sub-epidermal blister and IgG deposition at the basement membrane zone. Which of the following are not associated with this disorder:

- A immunodeficiency
- B diabetes
- C HLA-DQ
- D thyroid disease
- E T-cell lymphoma.

34 A. Ciclosporin

This patient has a history and histology consistent with pemphigus vulgaris a rare autoimmune bullous disorder. It presents with fragile superficial blisters of the skin and mucus membranes, in half of all patients the blistering is confined to the mouth. It often presents in late middle age and is more common in Jews and Indians. The blisters are superficial and do not scar but lesions may pigment. Pathologically there are circulating autoantibodies to desmoglein 3 a component of the desmosomes which attach keratinocyte cells together. Oral corticosteroids are considered first line therapy for pemphigus, second line agents include azathioprine, cyclophosphamide, mycophenolate mofetil, IVIG and rituximab.

Table 2.4 Types of pemphigus

Name	Clinical features
Pemphigus vulgaris	Commonest, superficial blisters and erosions of the skin and mucosal surfaces
Pemphigus vegetans	Flaccid blisters are replaced by fungoid vegetations and pustules
Pemphigus foliaceus	Scaly crusted erosions with an erythematous base, no mucosal involvement
Pemphigus erythematosis	Similar to pemphigus foliaceus, photosensitive, may have positive ANA

E T-cell lymphoma.

34 A 68-year-old woman presents with areas of oral ulceration and a widespread skin rash. She describes numerous thin blisters that appear on her skin and within her mouth and burst easily leaving raw erosions. A skin biopsy is taken and the patient is started on oral prednisolone 1 mg/kg/day. The skin biopsy shows subcorneal ulceration with prominent acantholysis. When the patient attends for review the prednisolone is not controlling the rash and it is decided to add in a second agent. Which of the following would not be appropriate:

- A ciclosporin
- B azathioprine
- C cyclophosphamide
- D mycophenolate mofetil
- E oxytetracycline.

34 A. Ciclosporin

This patient has a history and histology consistent with pemphigus vulgaris a rare autoimmune bullous disorder. It presents with fragile superficial blisters of the skin and mucus membranes, in half of all patients the blistering is confined to the mouth. It often presents in late middle age and is more common in Jews and Indians. The blisters are superficial and do not scar but lesions may pigment. Pathologically there are circulating autoantibodies to desmoglein 3 a component of the desmosomes which attach keratinocyte cells together. Oral corticosteroids are considered first line therapy for pemphigus, second line agents include azathioprine, cyclophosphamide, mycophenolate mofetil, IVIG and rituximab.

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35 A 30-year-old woman is referred by the obstetrics team with an itchy rash. She is in the third trimester of her first pregnancy. The patient describes a rash starting around her striae and spreading over much of her abdomen but sparing the umbilicus. On examination there is a mixture of urticated, erythematous plaques and papules on the abdomen with some excoriation. Blood tests have been undertaken by the obstetric team and show no abnormalities. What are the foetal and newborn risks for this woman's child:

35 D. No risk to foetus, no risk to newborn

This patient is presenting with pruritic urticarial papules and plaques of pregnancy (PUPPP), the most common skin complaint of late pregnancy. Patients present with itchy, urticated papules and plaques starting in the striae and sparing the umbilicus. It is more common in primiparous ladies and does not tend to recur in future pregnancies. Treatment is with potent topical steroids and antihistamines. The rash resolves spontaneously and rapidly after delivery and there is no risk to the foetus or newborn.

36 An 18-year-old girl presents as an emergency with a blistering rash, she is 33 weeks pregnant with her first child. She describes an abrupt onset of an itchy rash starting around the umbilicus and spreading. The next day numerous blisters started to appear from these itchy areas. On examination the patient has numerous intact blisters on an urticated base. She asks when the rash will resolve:

- A just prior to delivery
- B immediately post-delivery
- C a few weeks/months post-delivery
- D it will not resolve
- E it may resolve at any time prior to delivery.

36 C. A few weeks/months post-delivery

This patient has gestational pemphigoid which is also known as pemphigoid gestationis or herpes gestationis. It occurs in late pregnancy or immediately post-partum with an urticated pruritic rash that develops into vesiculobullae, rarely it may involve the mucosa. All patients have circulating anti-basement membrane zone autoantibodies and the histology and direct immunofluorescence are analogous to bullous pemphigoid. The rash often takes weeks to months to resolve post-delivery and there is a high risk of the rash developing in future pregnancies. The condition carries a risk of premature delivery and small babies. In addition 10% of babies will develop the rash transiently after delivery. Treatment is with oral prednisolone.

thankyou